

Parasoft C/C++ Test

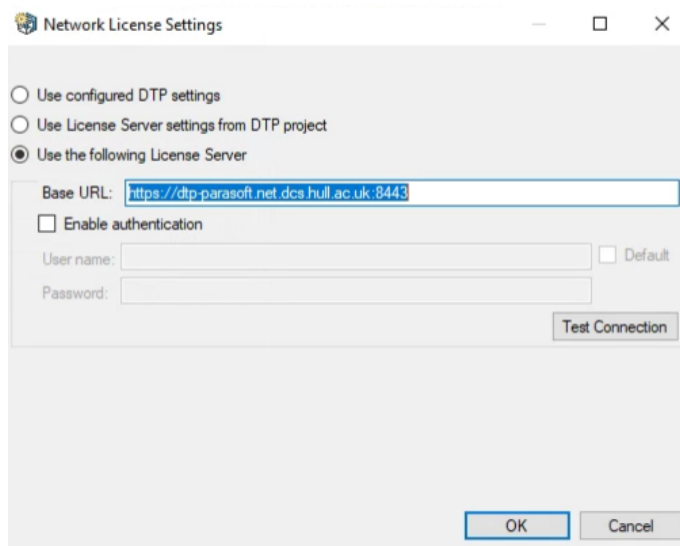
This tutorial introduces the reader to code verification using Parasoft.

Installing Parasoft C/C++ Test (This step is not required in a CS Lab)

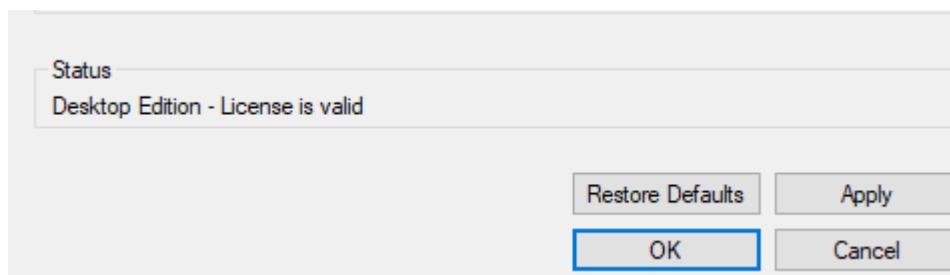
You MUST have Visual Studio Enterprise 2022 installed for Parasoft to work.

1. Close Visual Studio if you have it open.
2. Download Parasoft C/C++ Test from [Canvas](#).
3. Run the Parasoft installation and follow the instructions, using the recommended settings.
4. Restart Visual Studio
5. Select **Parasoft->Preferences**, then the **License** tab and finally **configure...**
6. Select **Use the following License Server** and then enter the Host name <https://dtp-parasoft.net.dcs.hull.ac.uk:8443> and hit **OK** then hit **Apply**

For this to work you MUST be connected to the University VPN using Global Connect



Parasoft should now be installed and licensed



Installing Parasoft C/C++ Test Rulesets

These are instructions for using installing the 700120 rule set in Parasoft C++ Test:

1. Open any Visual Studio solution.
2. Download the **Parasoft.zip** containing the rules from the module site on Canvas.
3. Unzip the rules files onto your OneDrive in a suitable location e.g.
C:\Users\csswjl\OneDrive - hull.ac.uk\Application Data\Parasoft
4. Select **Parasoft->Preferences**, then the **Configuration** tab
5. Replace the **user configurations** path with the location you just unzip the rules file e.g.
C:\Users\csswjl\OneDrive - hull.ac.uk\Application Data\Parasoft\cpptest\configs\user
6. Replace the **user rule map** path with the location you just unzip the rules file e.g.
C:\Users\csswjl\OneDrive - hull.ac.uk\Application Data\Parasoft\cpptest\rules\user
7. Then select the **Apply** button, and then **Ok**.
8. Select **Parasoft-> Test Configurations**
9. Expand the **User** tree, and you should see an entry **700120 rule set 2024**.
10. Double click this entry to active it
11. Right click this entry and **Set as Favourites -> 1 Recommended Rules**
12. Then select the **Apply** button, and then **Ok**.

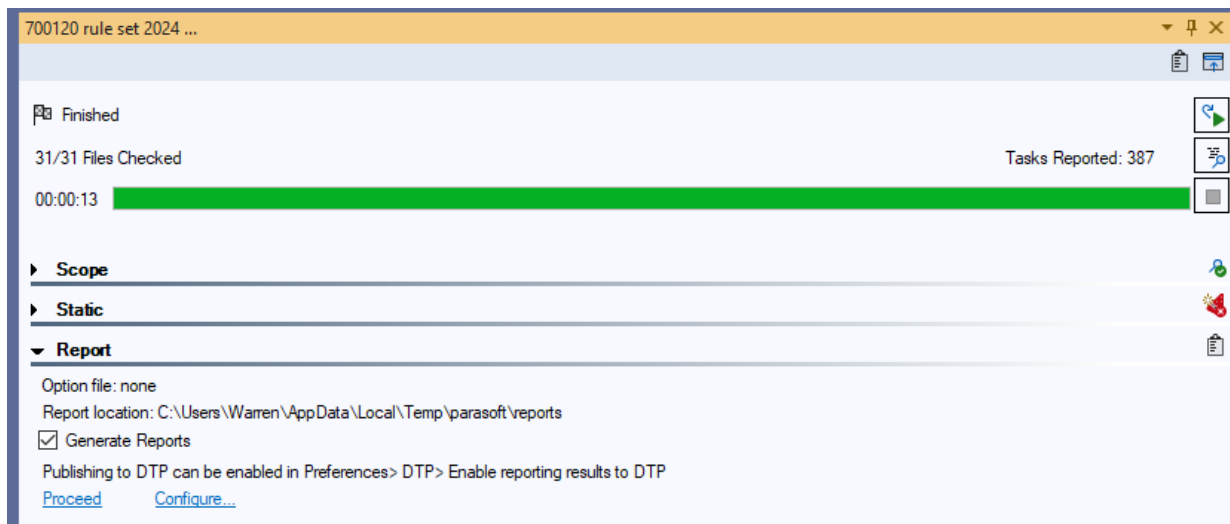
Using Parasoft C++ Test

These are instructions for using Parasoft C++ Test:

It is important to know that Parasoft will only validate any selected files in the Solution Explorer. Therefore, if you want Parasoft to validate all of your code, then select the project in the Solution Explorer before running the test. If you only want to run Parasoft on a selection of code, then select the files in the Solution Explorer before running the test.

Select **Parasoft -> Run "700120 rule set 2025"** from the main menu or from the Solution, Project or file context menu.

When the test is complete it will display a message



Ensure that **Generate reports** is ticked.

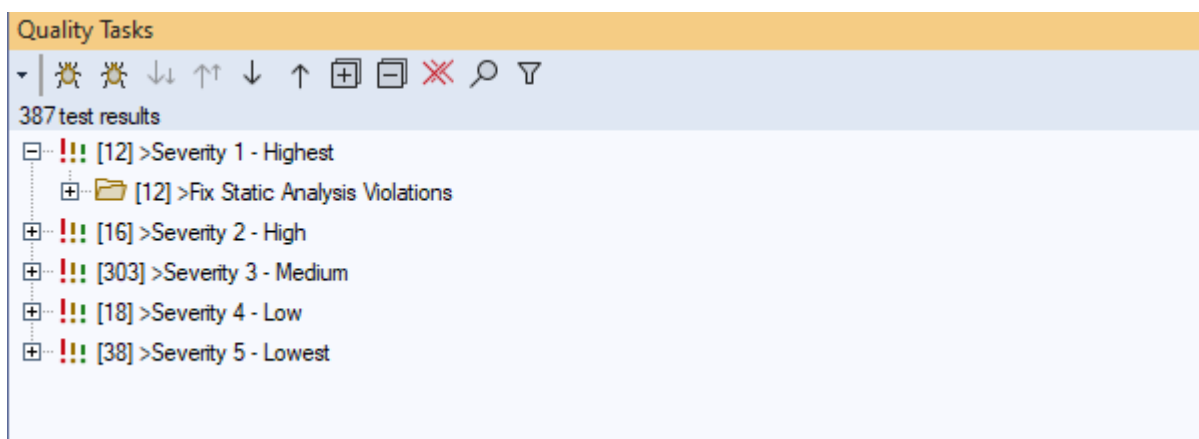
A report of the violations is displayed in the web browser

Click the **severity** link to sort by severity

All Findings by Category

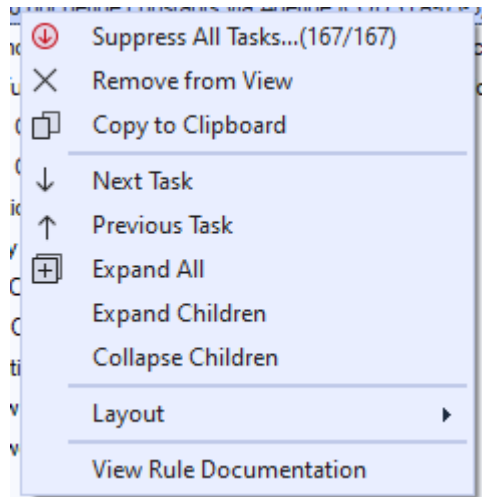
Category | [Severity](#)

A **Quality Tasks** window is also opened in Visual Studio



This can be sorted by severity, or any of the other column headings

Right click on an error and select **View Rule Documentation** to get a detailed description of the issue and how to fix it, displayed in a web browser.



Hint: This documentation is an invaluable resource, please make the most of it.

Using Parasoft C++ Test in an assessment

Save the violations report from the web browser, and submit as part of your assessment.